Aayush Kumar Chaudhary

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Research Interests: Image Processing; Compter Vision, Eye tracking; Machine Learning; Human Computer Interaction

Education

2016/21* **Ph.D. student, Chester F. Carlson Center for Imaging Science (Carlson CIS), Rochester Institute of Technology (RIT).** GPA: 3.79
Area of Focus: Eye Tracking Methodology (Image Processing & Machine Learning)
Relevant Courses: Image Processing and Computer Vision, AI Explorations, Deep Learning for Vision, Multiview Imaging, The Human Visual System, Fourier Methods for Imaging, Probability Noise and System Modeling

2010/14 **Undergraduate (B.E.), Central Campus, Pulchowk, Institute of Engineering (IOE), Tribhuvan University** in Electronics and Communication Engineering. (University Rank: 1 (Engineering Stream))
Thesis: Teleoperation of Mobile Robotic Arm Through Imitation of Human Arms Movement (Advisor: Dr. Sanjeeb P. Panday).

Research Publications

Journal Articles

1. **Chaudhary**, A. & Pelz, J. (2019, April). Motion tracking of iris features to detect small eye movements. *Journal of Eye Movement Research*, *12*(6). doi:10.16910/jemr.12.6.4

Conference Proceedings

1. **Aayush K. Chaudhary**, Kothari, R., Acharya, M., Dangi, S., Nair, N., Bailey, R., … Pelz, J. (2019). RITnet: Real-time Semantic Segmentation of the Eye for Gaze Tracking. In *The IEEE International Conference on Computer Vision (ICCV) Workshops (to appear)*.
2. **Chaudhary, Aayush K.** (2019). Motion tracking of iris features for eye tracking. In *Proceedings of the 11th acm symposium on eye tracking research & applications* (53:1–53:3). ETRA '19. Denver, Colorado: ACM. doi:10.1145/3314111.3322872
3. Nair, N., Kothari, R., **Aayush Chaudhary**, Pelz, J., Diaz, G., & Bailey, R. (2019). RIT-Eyes: rendering eye images under naturalistic conditions for eye-tracking applications. In *The IEEE International Conference on Computer Vision (ICCV) Workshops (to appear)*.

Skills

Programming Languages	Python, Matlab, Arduino, (Basic ability with C and C++)
Deep Learning Library	PyTorch (Basic ability with Keras and Tensorflow)
Eye Trackers	Pupil Labs Eye trackers, custom made trackers (designed eye tracker using a mirrorless digital camera)
Misc.	Circuit design tools (Kicad, PCB Wizard, Proteus), motor control (stepper, servo, DC motors), log extracting tools for chipsets (Qualcomm, Intel, Spreadtrum and Shannon), photography (iris images).

Experience

2017– **Research Assistant, Multidisciplinary Vision Research Laboratory** Carlson CIS.
Advisor: Prof. Jeff B. Pelz.
Research is focused on developing methods and algorithms for improving the precision of eye trackers.
Designing psychophysical experiment for human subjects in order to study various eye movements.
Study of feasibility of using features of iris for gaze estimation.

2016/17 **Teaching Assistant** Chester F. Carlson Center for Imaging Science
Courses: Vision and Psychophysics, Linear Methods of Imaging

2014/16 **Senior Engineer** Samsung India Electronics Pvt. Ltd (SIEL, Nepal branch)
Worked towards network performance testing and validation (METI team).
Helped to solve two critical market issues and reported numerous test issues.
My contribution in Q1'2016 was recognized with a performance award.

Awards

Professional Achievement

2016 **Performance Award in recognition of High Performance for Q1'2016**,Samsung India Electronics Pvt. Ltd., Nepal Office

Academic Achievement

2019 **Winner of Facebook OpenEDS competitionn.** Our model RITnet is memory efficient (0.98 MB) and achieves 95.3% accuracy on the 2019 OpenEDS Semantic Segmentation challenge.

2015 **NCell Excellence Award**, Received an award for being the overall topper of Department of Electronics And Communication, Pulchowk Campus.

2010-2014 **University Rank** 1st **(Engineering Stream)**, Rank First in All Bachelor's Degree programs of I.O.E. Tribhuvan University.

2014 **NCell Scholarship Award**, Third-year topper of Department of Electronics And Communication, Pulchowk Campus.

2010 **Nepal Board Topper**, Scored the highest score in the country in higher secondary (12th grade) Science faculty for the examination conducted by Higher Secondary Education Board (H.S.E.B.).

Merit Scholarship

2016– **RIT Ph.D. Merit Scholarship**, Financial Support for Ph.D. studies at RIT.

2010-2014 **College Fellowship**, Awarded for outstanding academic performance in each semester of Bachelor of Engineering by I.O.E., Central Campus, Pulchowk.

Full Academic Scholarship, Awarded to pursue B.E. at I.O.E., Central Campus, Pulchowk based on nationwide competitive examination.

Activities & Leadership

2014 **Coordinator, LOCUS 2014 (Technological Event)**, Coordinated and organized the 11th national technological festival- the biggest technological festival of the country for Department of Electronics and Computer Engineering and Department of Electrical Engineering.

Activities & Leadership (continued)

2013/14 **Secretary, Amateur Radio Club** A licensed Ham-Radio Operator and worked as a secretary for the that was involved in communication set up at different locations during the earthquake.

2013 **Team Member, ABU Robocon 2013, Vietnam**, Was an integral part of the hardware team (designing and building circuit boards) for the both manual and automatic robot that represented Nepal in the Asia-Pacific Broadcasting Union (ABU) Robocon 2013, Vietnam.

2011/12 **Secretary, Electronics Club** Organized crash course to freshmen regarding Electronics circuits and was involved in the launching of the first magazine of Electronics Club 'Graphene'.

Academic Projects

2019 **3D model of eye to predict gaze using Structure from Motion**

2018 **segmentation using Fully Convolutional Networks**

2017 **Emission Zone Gas Analysis System** Group project

Trajectory of Moving Object from Visual Cues

Sentimental Analysis of IMDb Movie Reviews

2014 **Teleoperation of Mobile Robotic Arm Through Imitation of Human Arms Movement (Undergraduate Thesis)**
RGB-D image processing on images acquired from Kinect camera to detect hand position. Electronics circuit board design: H-bridge for motor controls (stepper: robotic arm, starter motors: mobile base), interface circuits for microcontrollers, IMU sensor fusion to identify hand pose.

2013 **Spectrum Analyzer in FPGA (Field-programmable gate array)**
Implementation of 32-Point FFT on FPGA using VHDL

Miscellaneous

Poster Presentation

2019 **In 2019 Symposium on Eye Tracking Research and Applications (ETRA '19)**, Motion Tracking of Iris Features for Eye Tracking.

2018 **Graduate Showcase**, Motion Tracking of Iris Features to Detect Small Eye Movements.

Academic & Professional Service

2019 **Reviewer**, Journal of Eye Movement Research (JEMR).

2016 **Judge, LOCUS 2016 (Technological Event)**, Represented SIEL, Nepal branch as one of the main judge of the event Samsung Galaxy App Quest conducted to promote application development in Nepal.

Participation

2018 **IdeaLab**, Presented TUBBLE, an engaging Toy For Active Kids.

Miscellaneous (continued)

2014 **Child App Competition**, Presented a social interacting app 'Beautiful Minds' for autistic childen and was one of the finalist of the Child App Competition. It was organized by Microsoft Innovation Center (Nepal) and UNICEF Nepal.